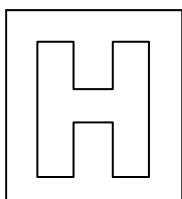


Candidate Name: _____

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2017 Preliminary Examination 2

Pre-University 3

H2 Biology**9648/01****Paper 1 Multiple Choice****21 September 2017****1 hour 15 min**Additional material: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

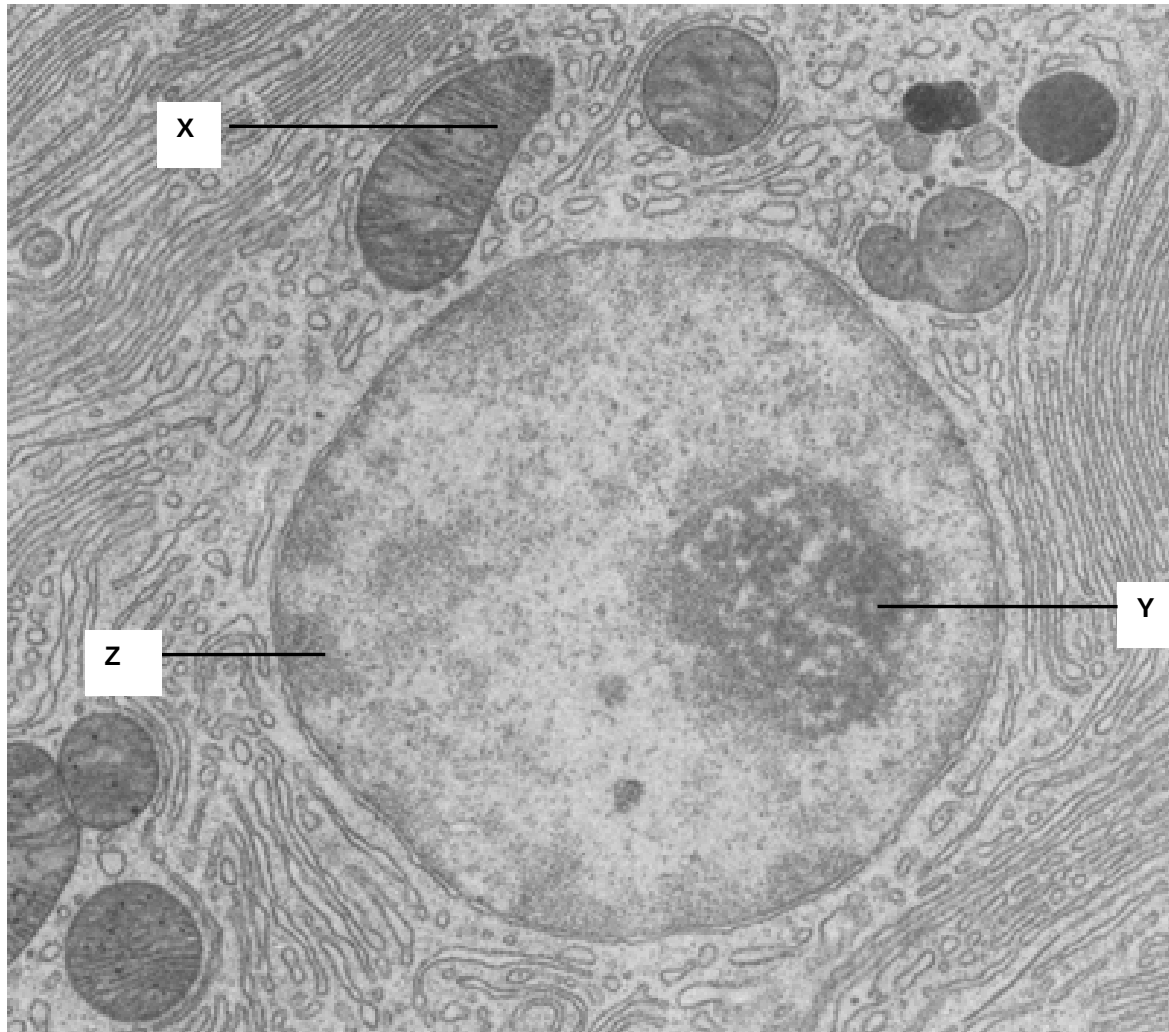
Write your name and Admission No. on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate

1. The electron micrograph below shows the structures found in a cell.

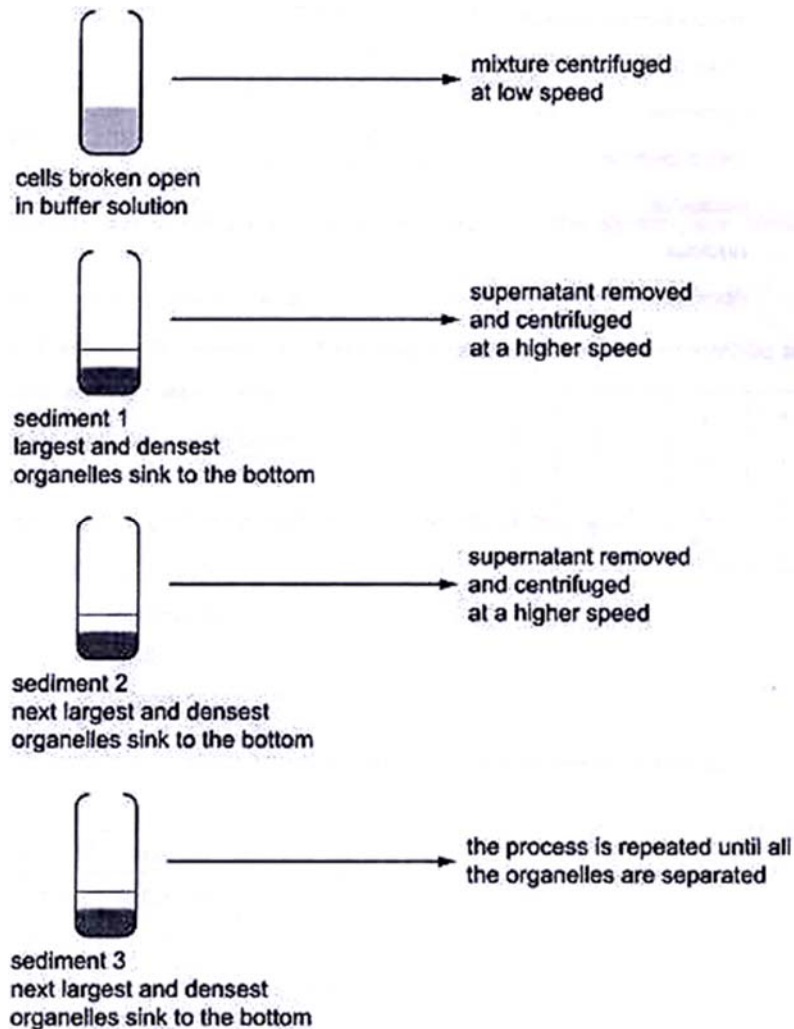


Which of the following statements is true for structures X, Y and Z?

	X	Y	Z
A	Contains reduced NAD^+ and reduced FAD	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD^+ and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Involved in oxidative phosphorylation	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

2. Fractionation is a process used to separate cell components according to their size and density.

The diagram shows the main stages in fractionation of a plant cell.

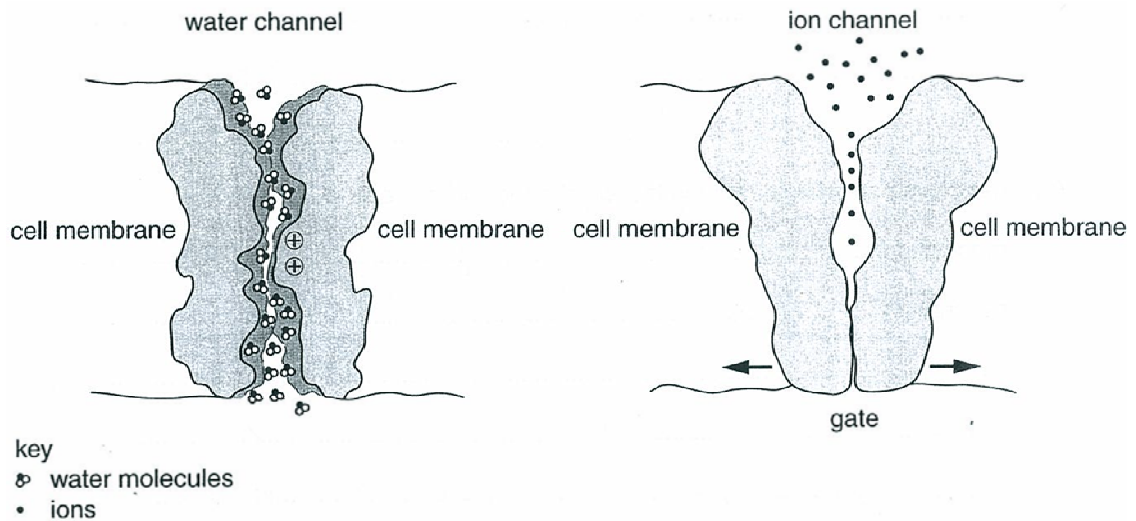


DCPIP and buffer solution (containing glucose, fructose, sodium bicarbonate) were added to each of the sediments, and the mixtures were left in the dark for fifteen minutes. Sediment 2 caused the DCPIP to be reduced.

Which organelle present in Sediment 2 caused reduction of DCPIP?

- A Chloroplast
- B Mitochondrion
- C Nucleus
- D Ribosome

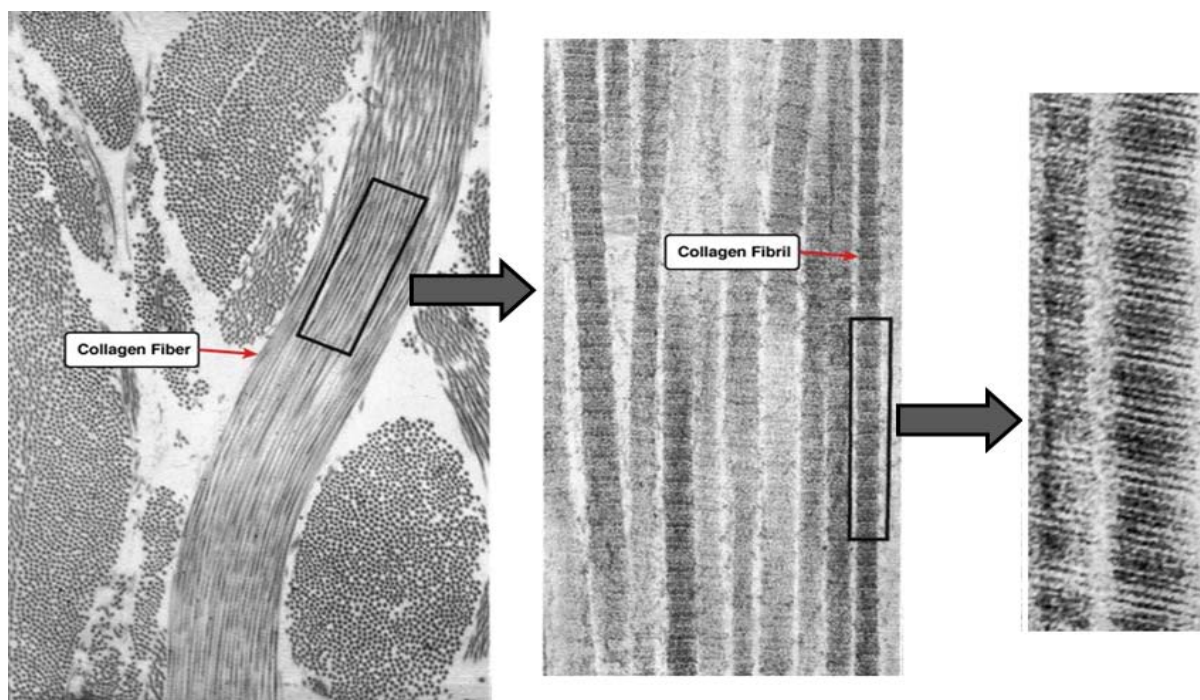
3. The figure shows water and ion channels that are found in the cell surface membrane of all cells.



Which of the following statements is true?

- A** Movement of water molecules through the water channel requires energy provided by the hydrolysis of ATP as water molecules are polar while the phospholipid bilayer of the membrane is hydrophobic.
- B** Common amino acid residues found on the protein surface surrounding the pores of both channels include valine and phenylalanine.
- C** Only the ion channel allows for the regulation of ion movement across the cell surface membrane.
- D** The ion channel is an example of a carrier protein as it is able to switch between two different conformations to allow the movement of ions.

4. Collagen is the main structural protein of the various connective tissues in animals. As the main component of connective tissue, it is the most abundant protein in mammal, making up from 25% to 35% of the whole-body protein content. The diagram below shows the structure of a collagen fibre and collagen fibrils.



Which of the following correctly accounts for the banded appearance of collagen fibril?

- A Intermolecular hydrogen bonds between polypeptide chains within tropocollagen.
 - B Covalent cross-linkages between tropocollagen chains.
 - C Staggered arrangement of collagen fibres.
 - D Sequence motif of Gly-X-Y where Gly is glycine, X is proline and Y is hydroxypoline or hydroxylysine.
5. Some foods contain 'hydrogenated vegetable oils'. These are unsaturated fats that have been converted to saturated fats.

Which property of the fats will have changed?

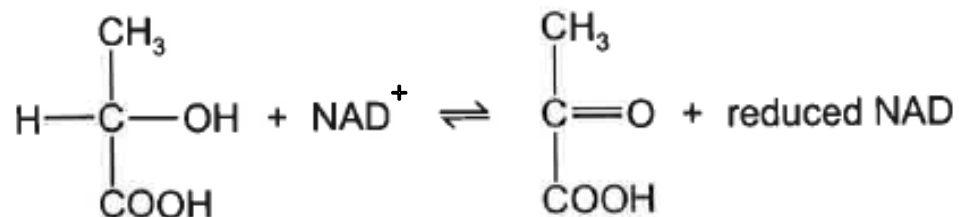
- A Their hydrocarbon chains will pack together more closely.
- B Their solubility in water will increase.
- C They will have more double bonds in their molecules.
- D They will remain liquid at room temperature.

6. Most wild plants contain toxins that deter animals from eating them. A scientist discovered that a toxin produced by a certain plant was also toxic to the same plant if it was applied to the roots of the plant. As the first step on finding out why the plant was not normally killed by its own toxin, he fractionated some plant cells and found that the toxin was in the fraction that contained the largest cell organelle. He also found that the toxin was no longer toxic after it was heated.

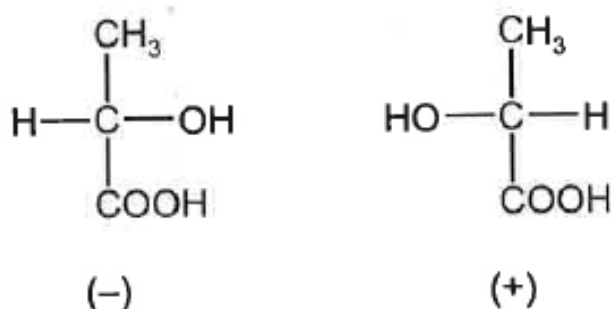
Which of the following statements are consistent with the scientist's observations?

- I The toxin was stored in the central vacuole.
 - II The toxin cannot cross the membrane of the organelle in which it is stored.
 - III The toxin was stored in chloroplast.
 - IV The toxin is likely to be lipid-soluble.
 - V The toxin may be an enzyme.
-
- A I, II and V
 - B I, IV and V
 - C II, III and IV
 - D III, IV and V

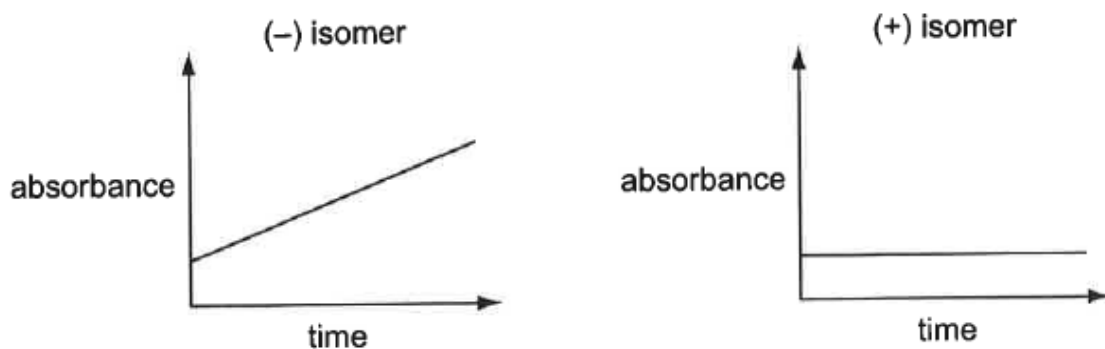
7. Lactic dehydrogenase catalyses the conversion of lactic acid as shown in the following equation.



Two forms (isomers) of lactic acid exist, (-) and (+), as shown below.



Reduced NAD absorbs ultraviolet light. NAD^+ does not. The activity of bacterial lactic dehydrogenase on two different isomers of lactic acid was compared. The absorbance of ultraviolet light was measured using an ultraviolet spectrophotometer. The graphs show the results.



What can be concluded about bacterial lactic dehydrogenase?

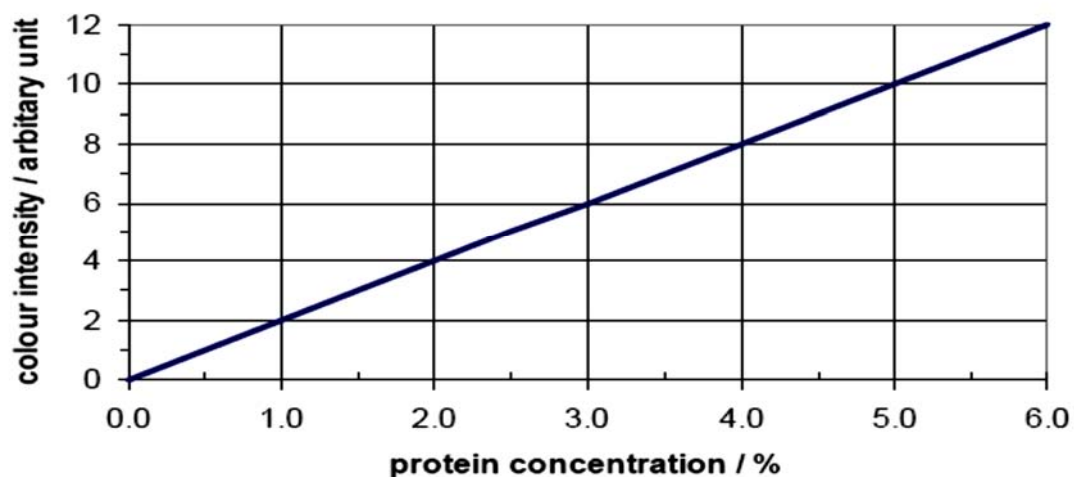
- A Molecules of both isomers fit the active site.
- B Molecules of neither isomer fit the active site.
- C The enzyme is specific to the (-) isomer.
- D The enzyme is specific to the (+) isomer.

8. In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in the table below.

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit			
	Tube A	Tube B	Tube C	Tube D
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + heated raw bean extract	Protease + unheated protein substrate + raw bean extract
0	10	10	10	10
30	4	6	7	9

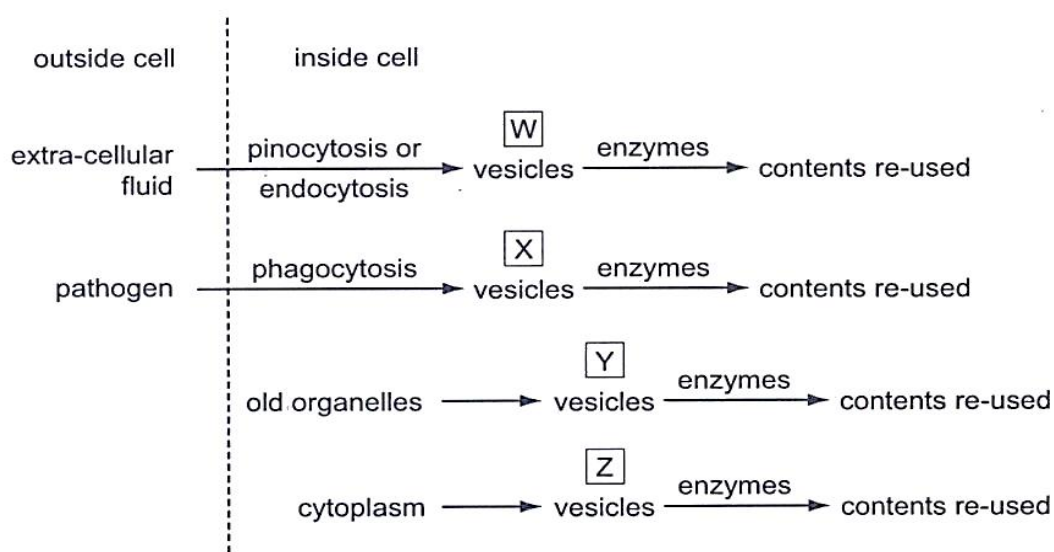
The standard graph obtained by using colorimetric method for determining concentration of protein solutions is shown below.



Which of the following combinations is correct?

	Test tube	Decrease in protein concentration / %
A	Tube A	4.0
B	Tube B	2.0
C	Tube C	3.0
D	Tube D	4.5

9. The flow chart shows processes which takes place inside animal cells.



Which processes require the activity of lysosomes?

- A W and X only
 B X and Y only
 C Y and Z only
 D All of the above
10. A student obtained a sample of DNA molecule. mRNA was transcribed from this DNA molecule. He then separated the two strands of the DNA sample by adding sodium hydroxide. The base compositions of each strand, that of the mRNA and a foreign DNA strand were analysed. The results of the analysis are shown in the table below.

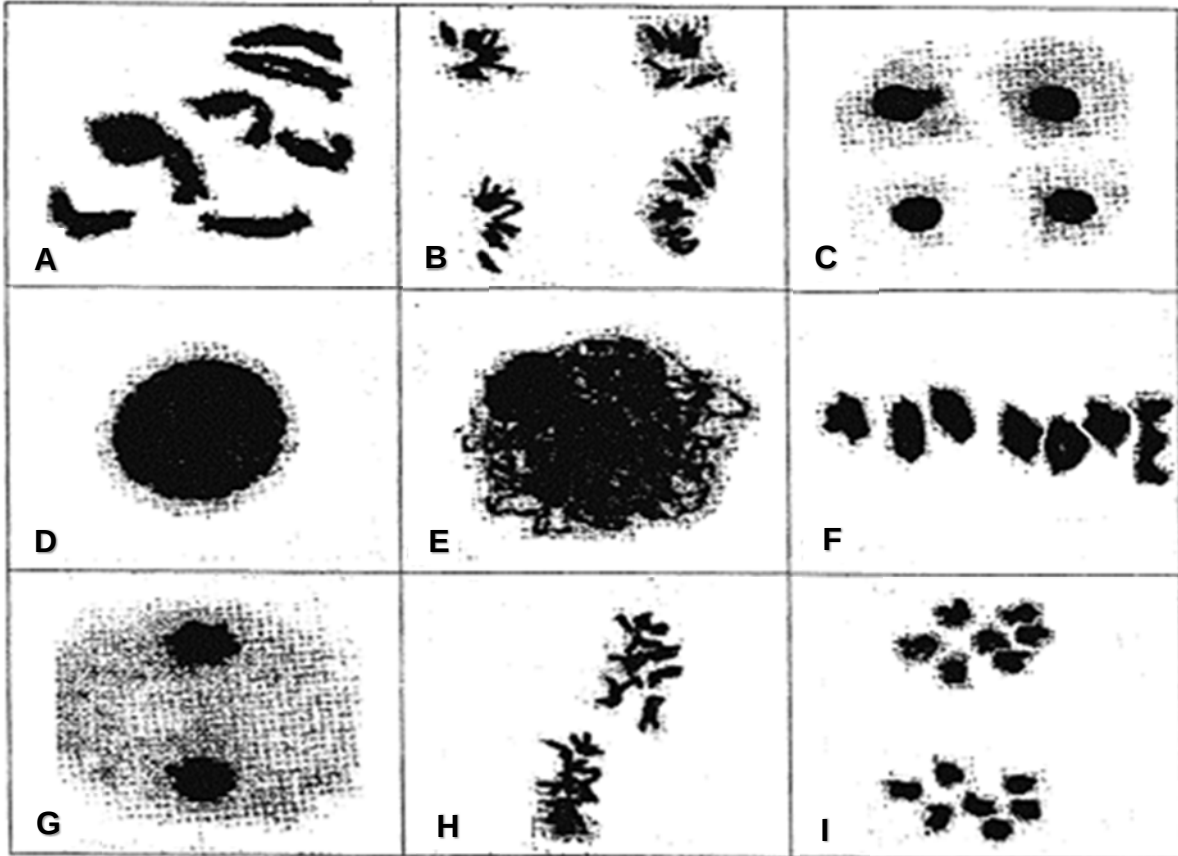
	A	G	C	T	U
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

Which strand of DNA serves as a template for mRNA synthesis?

- A Strand 1
 B Strand 2
 C Strand 3
 D Strand 2 and 3

Use the diagram below to answer Questions 11 and 12

The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant.



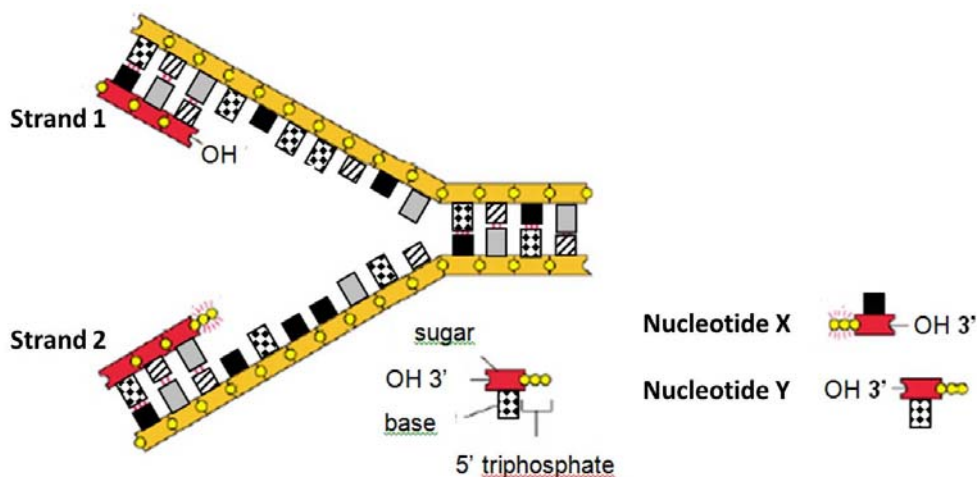
11. Which of the following combinations is the correct arrangement of letters in accordance with the chronological sequence of events for the above nuclear division process?

A	D	E	B	C	I	G	H	A	F
B	E	D	A	H	I	F	G	B	C
C	D	E	A	F	I	G	H	B	C
D	C	B	H	G	I	F	A	E	D

12. With reference to micrograph F, which of the following combinations is correct?

	Number of sets of chromosomes	Number of centromeres	Number of chromatids	Number of DNA strands
A	2	14	28	56
B	7	7	14	28
C	2	7	28	28
D	7	14	14	56

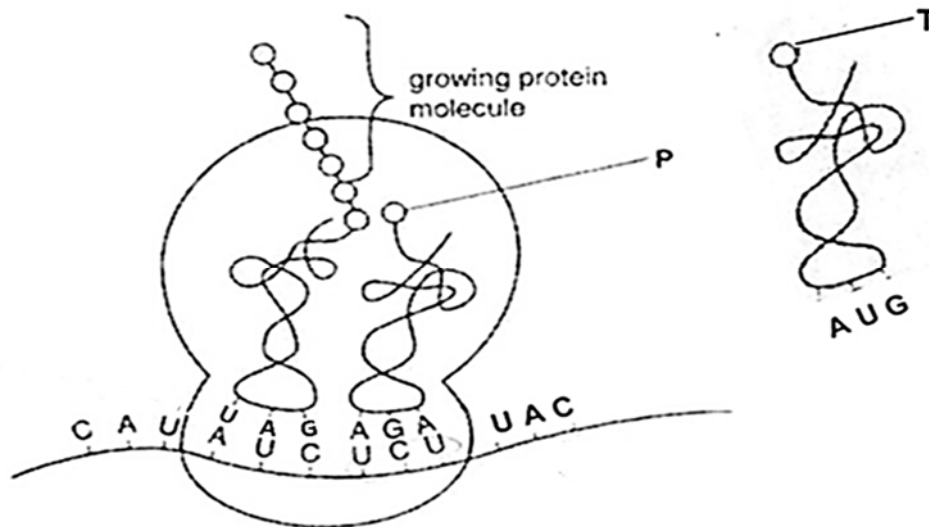
13. DNA replication is illustrated in the figure below.



Which of the following correctly describes the addition of the next nucleotide(s) to the DNA strands undergoing replication?

- A Nucleotide X will be added to the leading strand, which is strand 1.
- B Nucleotide Y will be added to the leading strand, which is strand 1.
- C Nucleotide X will be added to the lagging strand, which is strand 2.
- D Nucleotide Y will be added to the leading strand, which is strand 2.

14. The figure below shows a diagram of a ribosome bound to a mRNA strand during translation.



		Second Base				
		U	C	A	G	
First Base	U	phenylalanine	serine	tyrosine	cysteine	U
		phenylalanine	serine	tyrosine	cysteine	C
		leucine	serine	(stop)	(stop)	A
		leucine	serine	(stop)	tryptophan	G
	C	leucine	proline	histidine	arginine	U
		leucine	proline	histidine	arginine	C
		leucine	proline	glutamine	arginine	A
		leucine	proline	glutamine	arginine	G
	A	isoleucine	threonine	asparagine	serine	U
		isoleucine	threonine	asparagine	serine	C
		isoleucine	threonine	lysine	arginine	A
		(start) methionine	threonine	lysine	arginine	G
	G	valine	alanine	aspartate	glycine	U
		valine	alanine	aspartate	glycine	C
		valine	alanine	glutamate	glycine	A
		valine	alanine	glutamate	glycine	G

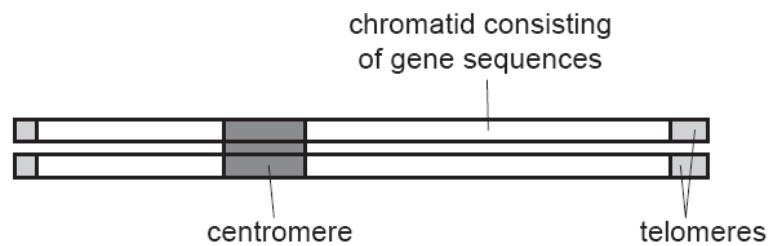
Using the codon table provided, which of the following options correctly identifies amino acids **P** and **T**?

	P	T
A	Serine	Histidine
B	Serine	Tyrosine
C	Arginine	Methionine
D	Arginine	Leucine

15. The ends of eukaryotic chromosome contain a special sequence of DNA called a telomere. Human telomeres consist of repeating TTAGGG sequences which extend from the ends of the chromosomal DNA.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in a shortening of the telomeres.

The diagram shows a eukaryotic chromosome.



What is a consequence of the loss of repeating DNA sequences from the telomeres?

- A The cell will begin the synthesis of different proteins.
- B The cell will begin to differentiate as a result of the altered DNA.
- C The number of mitotic divisions the cell can make will be limited.
- D The production of mRNA will be reduced.

16. The table shows a comparison of some aspects of the genomes and protein-coding genes between the prokaryote *Escherichia coli* and the eukaryote fungus *Saccharomyces cerevisiae*.

	<i>E.coli</i>	<i>S.cerevisiae</i>
Genome length/base pairs	4 640 000	12 068 000
Number of protein-coding genes	4300	5800
Proteins with roles in:		
Metabolism	650	650
Energy release/storage	240	175
Membrane transport	280	250
Transcription	240	400
Translation	180	350
Cell structure	180	250

What could not account for the differences in the number of protein-coding genes?

- A Many catabolic pathways for using carbon compounds in prokaryotes.
- B The presence of introns in the DNA of eukaryotes.
- C The presence of membrane-bound organelles in eukaryotes.
- D The use of histones to package DNA in eukaryotes.

17. The following are characteristics of eukaryote transcription.

- Promoters are activated by transcription factors that recognise specific DNA sequences and other sequences that are very similar.
- Within a promoter, there may be recognition sites for more than one transcription factor.
- Similar specific DNA sequences can be recognised by more than one transcription factor.
- Each transcription factor may be capable of recognising a number of promoter recognition sites.

What explains the different levels of expression of a eukaryotic gene?

- A** Competition between recognition sites present in the promoter for transcription factors.
- B** Competition between transcription factors that recognise the same sites of a promoter.
- C** The number of transcription factors that recognise the same sites of a promoter.
- D** The number of different types of transcription factors.

18. The figure below shows a human karyotype.



What can be concluded from the karyotype provided?

- A There was non-disjunction during meiosis I in the mother.
- B There was non-disjunction during meiosis II in the father.
- C One contributory gamete to the zygote is an egg with an X and a Y chromosome.
- D One contributory gamete to the zygote is a sperm containing two X chromosomes.

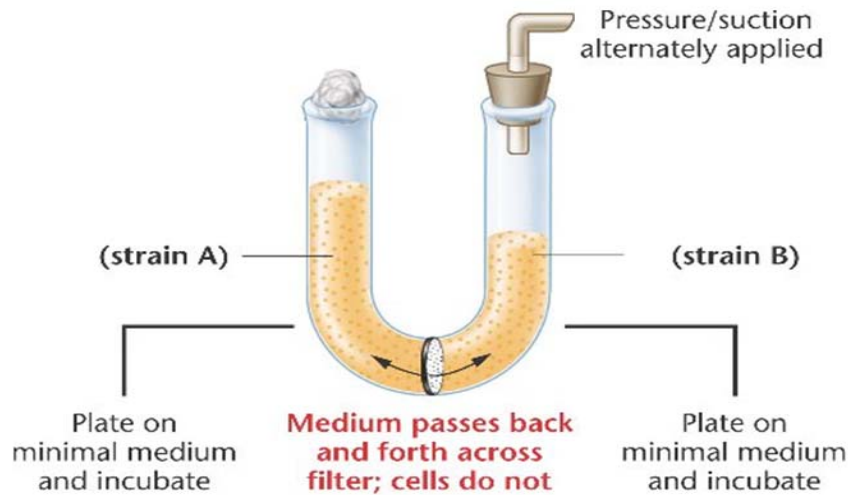
19. Below are some statements related to cancer:

- I. Oncogenes can be detected by introducing fragmented DNA from cancer cells into suitable cell lines and isolating colonies that display cancerous properties.
- II. Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies.
- III. Viruses and other infectious agents play no role in human cancers.
- IV. In the cellular regulatory pathways that control cell growth and proliferation, the products of oncogenes are inhibitory components and the products of tumour suppressor genes are stimulatory components.
- V. When analysed, cancer cells are often found to have only one mutation in a regulatory pathway that controls cell proliferation.

Which of the following statements are true?

- A I and II only.
- B I, II and III only.
- C I, III and V only.
- D I, II, IV and V only.

20. Bacteria can undergo genetic recombination, a process by which genetic information from one bacterium is transferred to, and then recombined with, that of another bacterium.



The Davis U-tube, shown above is an apparatus used to investigate possible genetic recombination between bacteria. In the experiment, researchers placed *Salmonella typhimurium* **strains A** and **B** in the U-tube separated by a filter, thus preventing direct cell contact but allowing growth to occur in a common medium. When samples were removed from both sides of the filter, recombinants (containing genetic material from both **strain A** and **B**) were recovered only from the side of the tube containing **strain A** bacteria. Researchers postulated that a filterable agent was released by the **strain B** cells and was responsible for transferring the new genetic information.

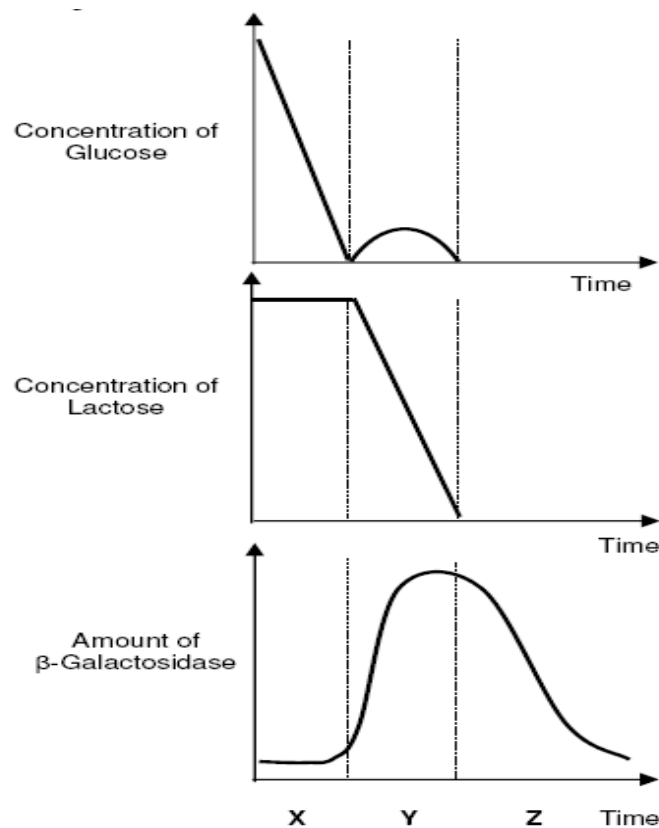
Three subsequent observations were useful in identifying the filterable agent:

1. The filterable agent was released by the **strain B** cells only when they were grown in association with **strain A** cells.
2. The addition of DNase, which enzymatically digests naked DNA, did not render the filterable agent ineffective.
3. The filterable agent could not pass across the filter of the Davis U-tube when the pore size was reduced below the size of bacteriophages.

Which process has occurred?

- A Transduction
- B Conjugation
- C Transformation
- D Binary fission

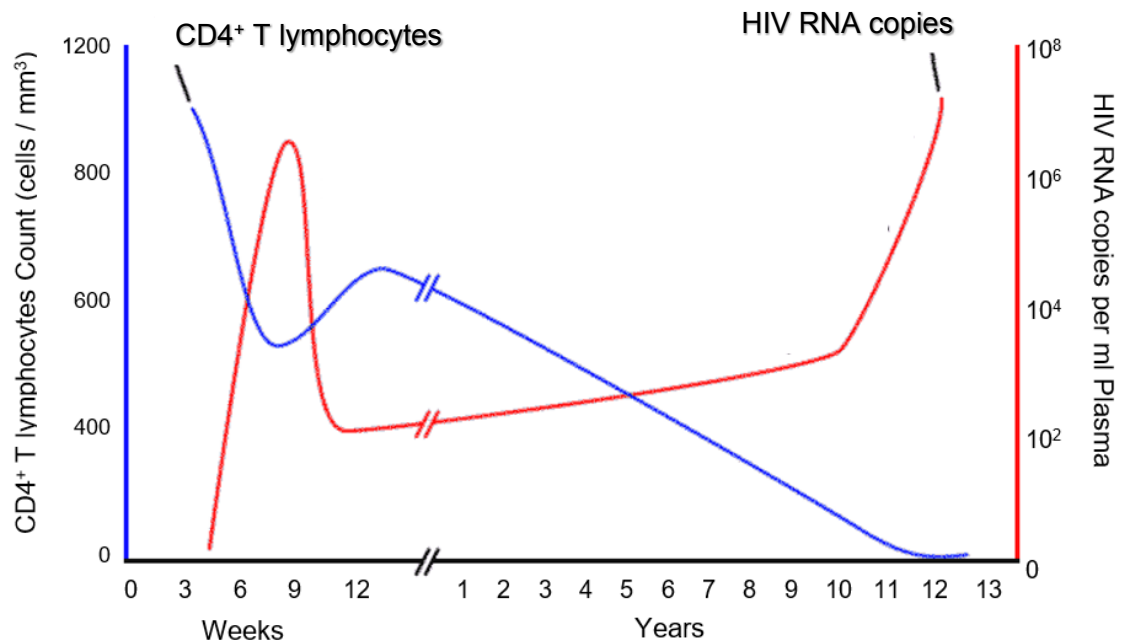
21. An experiment was conducted to examine the effects of glucose and lactose on the levels of β -galactosidase in *E.coli*. Lactose and glucose were added to a culture of bacteria at the start of the experiment and the levels of each were measured at specific time intervals. The results are shown in graphs below.



Which of the following statements could possibly account for period X to Z?

- A Binding of cAMP to the CAP-binding site enhances binding of RNA polymerase to the promoter for gene transcription in period Y.
- B Allolactose binds to the lac repressor, allowing it to assume an active configuration such that it can bind to the operator in period X.
- C CAP is inactive and disengages from the CAP binding site, hence increasing the affinity of RNA polymerase to the promoter for gene transcription in period Y.
- D mRNA of β -galactosidase has been degraded by nucleases in period Z.

22. The graph below shows HIV copies and CD4⁺ T lymphocytes counts over the course of a typical HIV infection.



Which of the following statements are false about how HIV infects the cell?

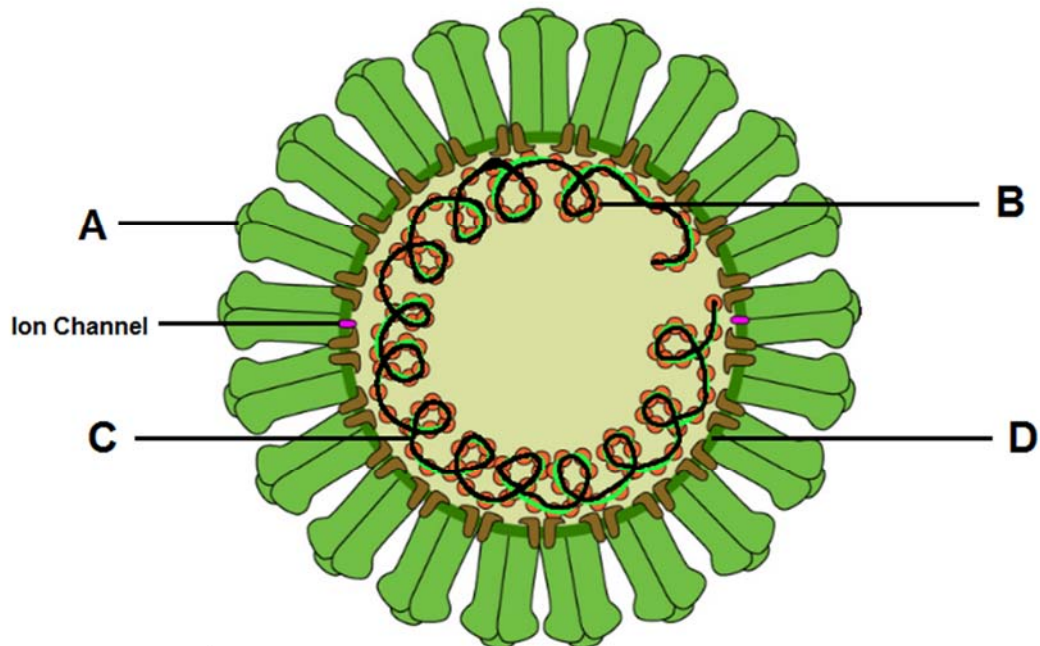
- I Complementary binding of the gp120 to specific CD4⁺ receptors on the T cells and HIV enters the host cell via receptor-mediated endocytosis.
- II RNA released into cytoplasm where reverse transcriptase uses negative-sense viral RNA as a template to synthesise a strand of cDNA and then form a double stranded viral DNA.
- III The DNA enters the nucleus and ligase catalyses the integration into the chromosome DNA to form a provirus.
- IV The provirus DNA is transcribed to form viral mRNA which are used as a template for translation of viral proteins such as nucleocapsids, viral envelope and viral enzymes.
- V Neuraminidase cleaves the long chains of polyproteins when newly assembled HIV bud out of host cells.

- A I and III only
- B I, II and III only
- C I, IV and V only
- D All of the above

23. Middle East respiratory syndrome (MERS) is a viral respiratory illness that was first reported in Saudi Arabia in 2012. Symptoms may range from fever, cough to shortness of breath.

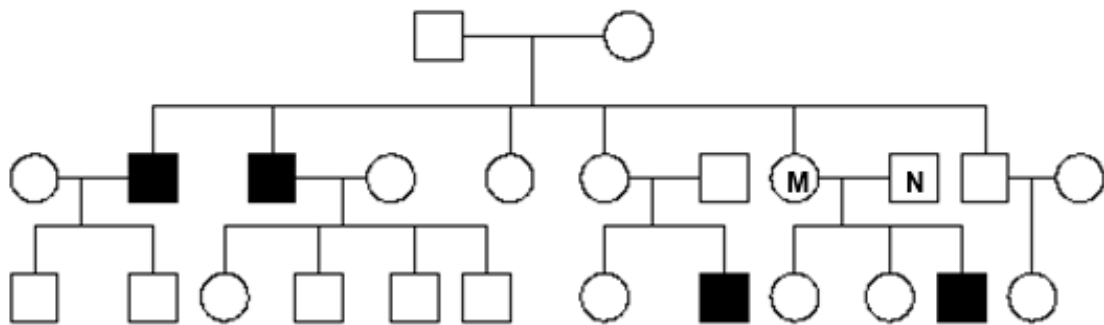
This infection is caused by the MERS-coronavirus (MERS-CoV) shown in the diagram below.

Which of the following components of MERS-CoV is not present in *Escherichia coli* bacterium?



Use the diagram below to answer Questions 24 and 25.

Hunter's syndrome is a serious genetic disorder. It interferes with the body's ability to break down and recycle specific mucopolysaccharides, also known as glycosaminoglycans or GAG. The visible signs and symptoms of Hunter syndrome in younger people are usually the first clues leading to a diagnosis. In general, the time of diagnosis usually occurs from about 2 to 4 years of age.



24. With reference to the pedigree diagram, which of the following is the correct mode of inheritance for Hunter's syndrome?

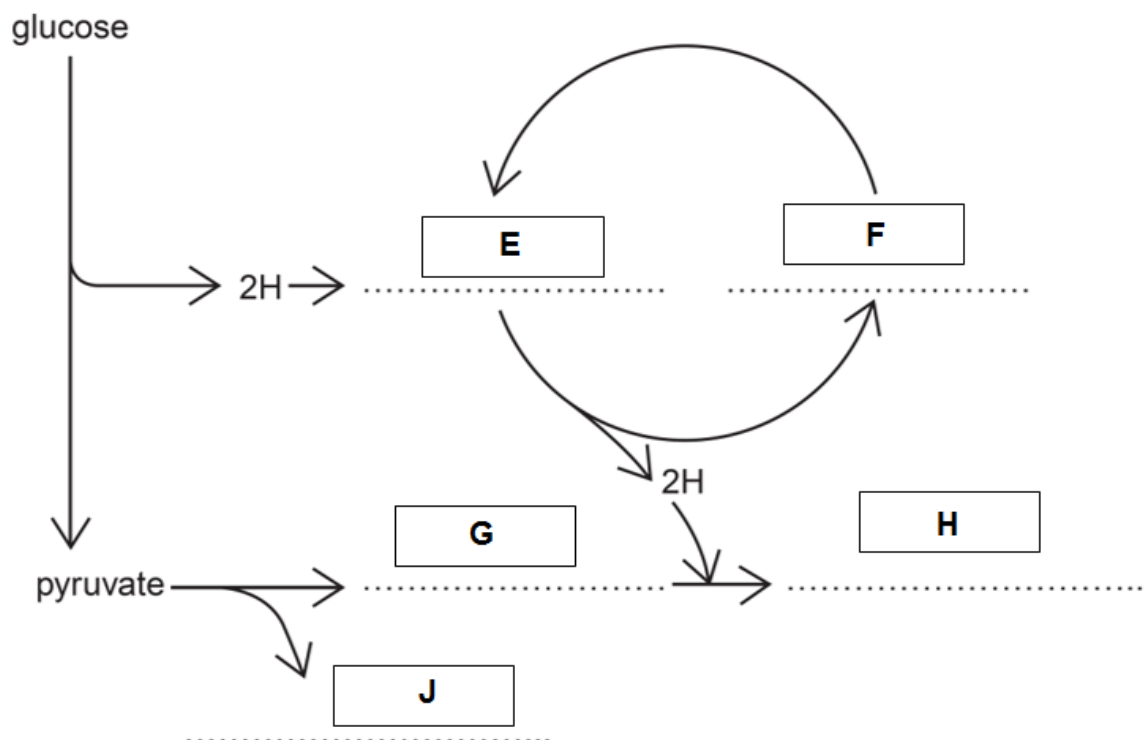
A Autosomal recessive
 B Incomplete dominance
 C Sex-linked recessive
 D Codominance

25. Mariah (**M**) married Nick (**N**) and had three children. One of the children had Hunter's syndrome.

What is the probability of their next child being an affected son?

A 0.5
 B 0.375
 C 0.25
 D 0.125

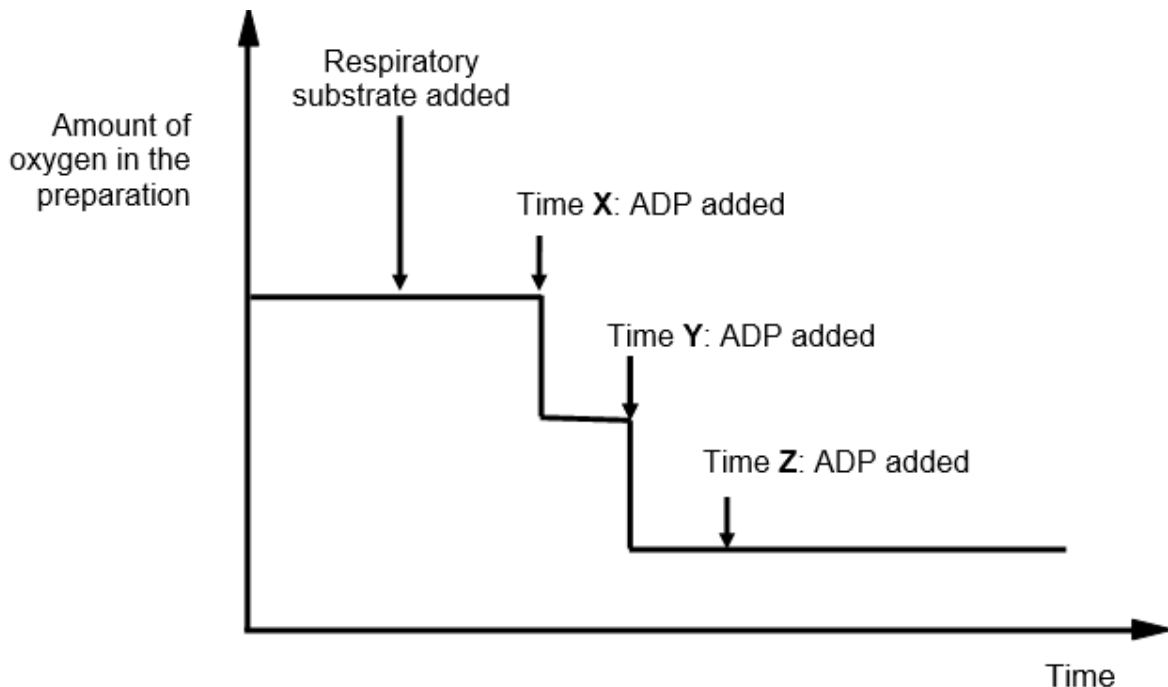
26. The figure below outlines a process that occurs in plant cells.



Which of the following combinations is correct?

	E	F	G	H	J
A	NADH	NAD ⁺	Ethanal	Ethanol	CO ₂
B	NAD ⁺	NADH	Ethanol	Ethanal	CO ₂
C	NADPH	NADP ⁺	Lactate	Lactose	O ₂
D	NADP ⁺	NADPH	Lactate	Lactose	O ₂

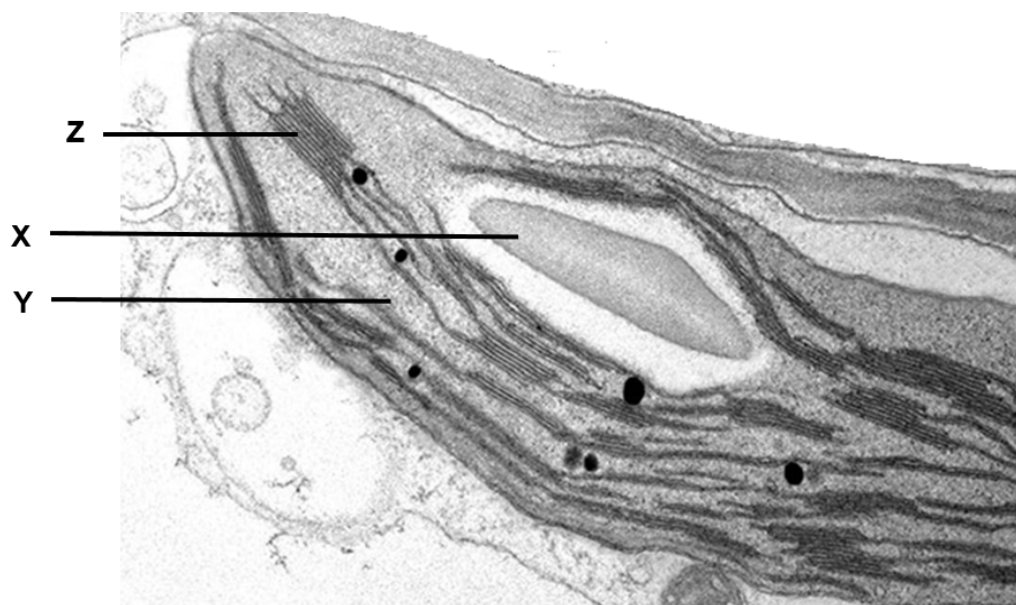
27. A suspension of mitochondria was isolated from liver tissue. Various substances were added to the suspension at different time intervals and the amount of oxygen remaining in the preparation was monitored over some time. The graph below shows the results as well as the times at which different substances were added.



Which of the following statement(s) could possibly be true?

- I Glucose is the respiratory substrate added.
 - II Between X and Y, oxidative phosphorylation occurred and oxygen acted as the final electron acceptor.
 - III Between Y and Z, chemiosmosis occurred where ATP synthase utilizes the proton-motive force to phosphorylate ADP to form ATP.
 - IV After Z, anaerobic respiration occurred as oxygen levels did not decrease even though ADP is added.
 - V After Z, inorganic phosphates, NADH and FADH_2 have been depleted.
- A** I, III and IV only
B II, IV and V only
C II, III and V only
D All of the above

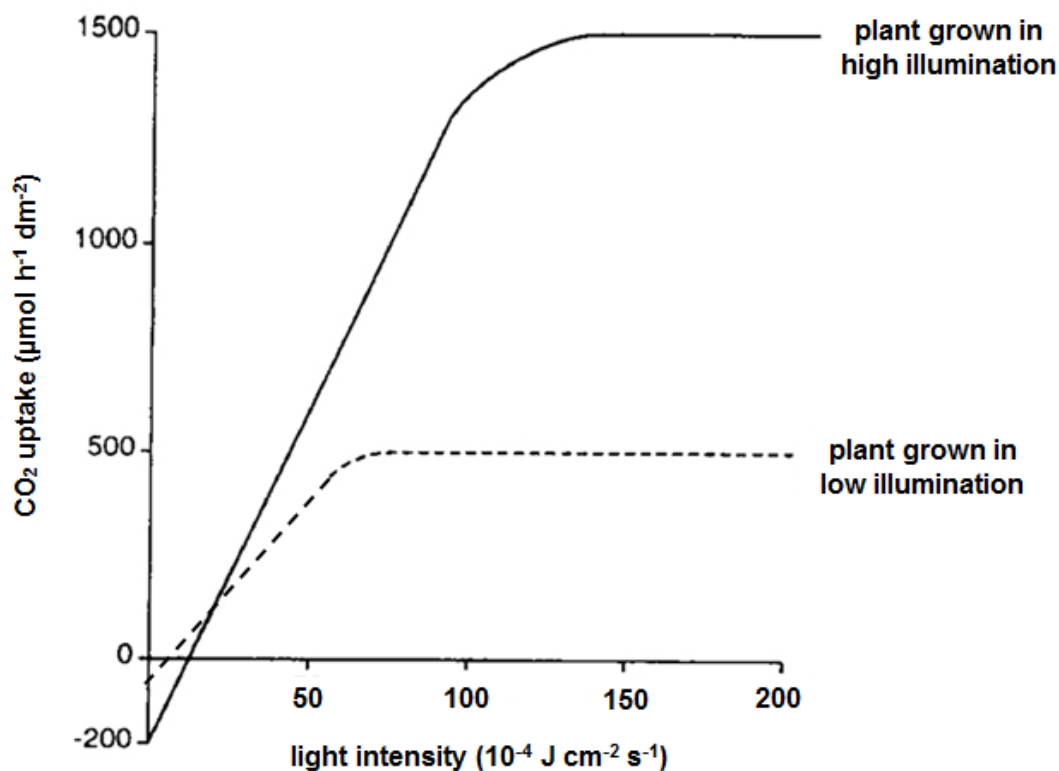
28. The diagram below is a transmission electron micrograph of a labelled organelle.



Which of the following combinations is correct?

	X	Y	Z
A	Consists of more amylopectin branches than amylose chains.	Contains low concentrations of protons and has an alkaline pH.	Site of non-cyclic and cyclic photophosphorylation.
B	Large central vacuole surrounded by a tonoplast and contains cell sap.	Site of Calvin cycle processes of carbon fixation, reduction and RuBP regeneration.	Site of ATP and NADPH synthesis.
C	Insoluble in water as hydroxyl groups are projected inwards into helical structures and unable to form hydrogen bonds with water.	Site of oxidation of NADPH to form NADP ⁺ as well as expenditure of ATP.	Photolysis of water occurs to generate protons, oxygen and electrons.
D	Consists of monomers joined together by $\beta(1\rightarrow4)$ and $\beta(1\rightarrow6)$ glycosidic bonds.	Consists of chlorophyll pigments with photosystems to facilitate light-dependent reactions.	Consists of cristae that increases surface area to volume ratio for more efficient ATP production.

29. Two groups of white mustard plants, *Sinapis alba*, were grown, one group under high illumination, the other under low illumination. When fully grown, the effect of increasing light intensity on the rate of photosynthesis in the two groups of plants was measured.



Which of the following statements can be concluded from the graph?

- A Below the compensation point, plants grown at high illumination give out less carbon dioxide than plants grown in low illumination.
- B The compensation point for plants grown in high illumination occurs at a lower light intensity than those grown in low illumination.
- C Light intensity is no longer a limiting factor for photosynthesis for light intensity above $150 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ for plants grown in high illumination.
- D For light intensity from $20 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$ to $50 \times 10^{-4} \text{ J cm}^{-2} \text{ s}^{-1}$, carbon fixation for the plants grown in high illumination is similar to that grown in low illumination.

30. Four proteins isolated from a human cell were investigated for their involvement in cell signalling pathways.

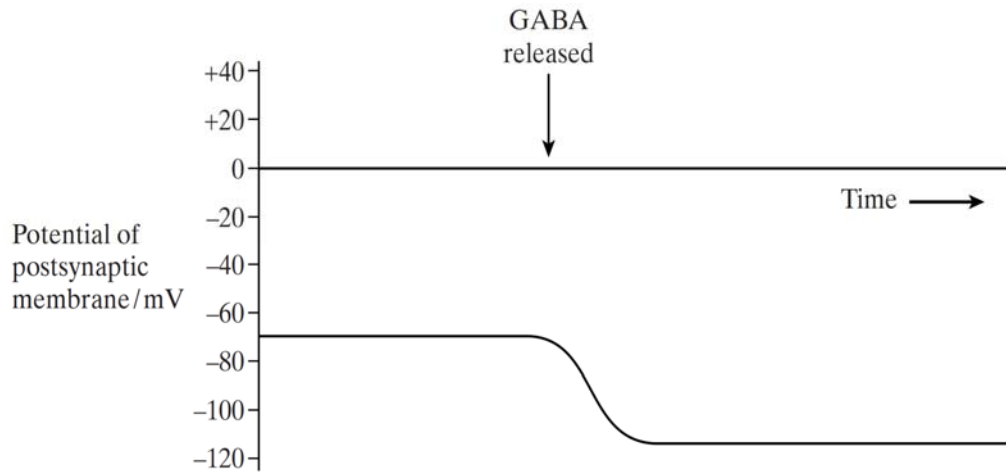
	Protein A	Protein B	Protein C	Protein D
Transmembrane domain	-	+	+	-
DNA binding domain	-	-	-	+
Enzymatic domain	+	-	+	-

Key: (+) = present , (-) = absent

Which of the following shows the correct identity of these four proteins?

	Protein A	Protein B	Protein C	Protein D
A	GPCR	Ras protein	RTK	Testosterone receptor
B	Ras protein	RTK	GPCR	Testosterone receptor
C	Testosterone receptor	GPCR	RTK	Ras protein
D	Ras protein	GPCR	RTK	Testosterone receptor

31. GABA is a neurotransmitter which inhibits the production of action potential. The figure below shows how the release of GABA from a pre-synaptic neurone affects the membrane potential of a post-synaptic membrane.



Which of the following options correctly explains why an action potential is less likely to occur if GABA is released?

- A GABA opens ligand-gated K^+ ion channels in the post-synaptic membrane, allowing K^+ to diffuse out of post-synaptic neuron, causing hyperpolarization.
 - B GABA closes voltage-gated Na^+ ion channels in the pre-synaptic membrane, allowing K^+ to diffuse out of pre-synaptic neuron, causing repolarization.
 - C GABA opens voltage-gated K^+ ion channels in the post-synaptic membrane, allowing K^+ to diffuse into the post-synaptic neuron, causing repolarization.
 - D GABA opens voltage-gated Na^+ ion channels in the post-synaptic membrane, allowing Na^+ to diffuse out of post-synaptic neuron, causing hyperpolarization.
32. The resting potential of a nerve axon is essential for action potential generation.

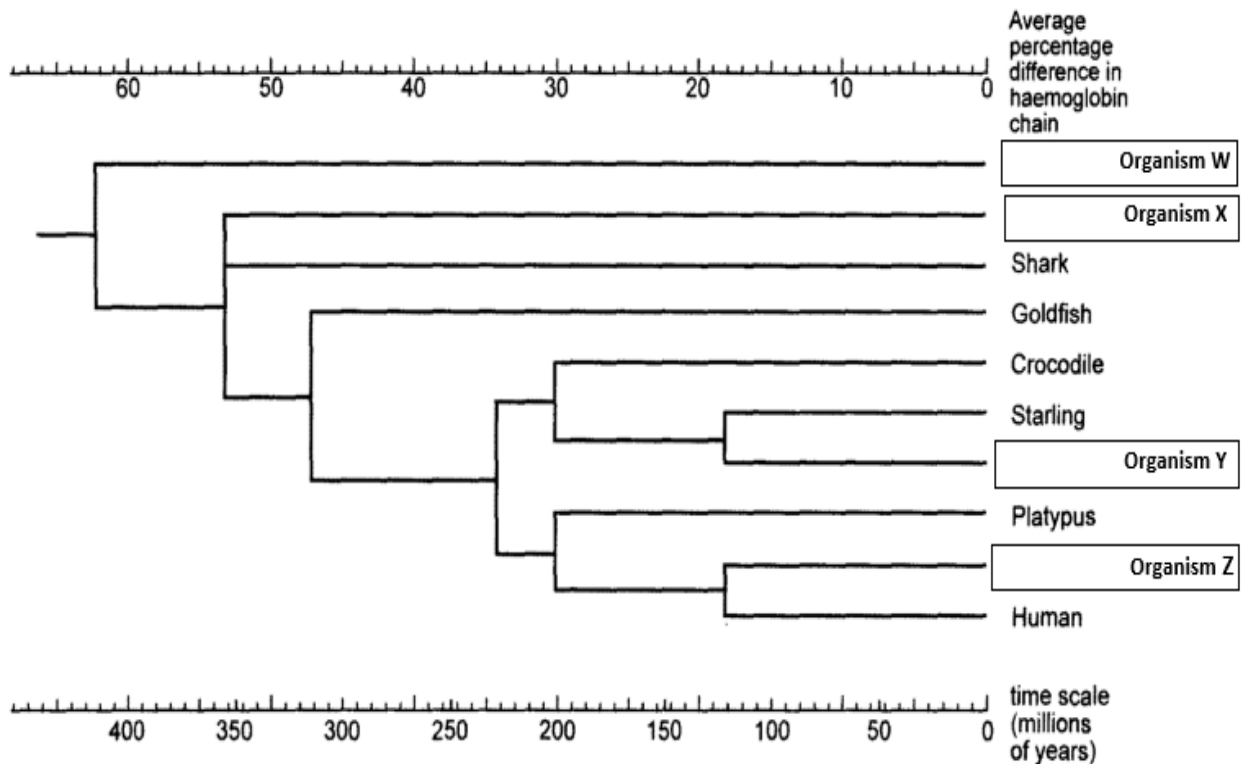
Which of the following, when instantaneously removed, would most rapidly bring the resting potential of a nerve axon close to 0 mV?

- A Active transport of K^+ ions into the cell
- B Active transport of Na^+ ions out of the cell
- C High membrane permeability to Na^+ ions
- D High membrane permeability to K^+ ions

33. The table shows the amino acid differences in the cytochrome b protein between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

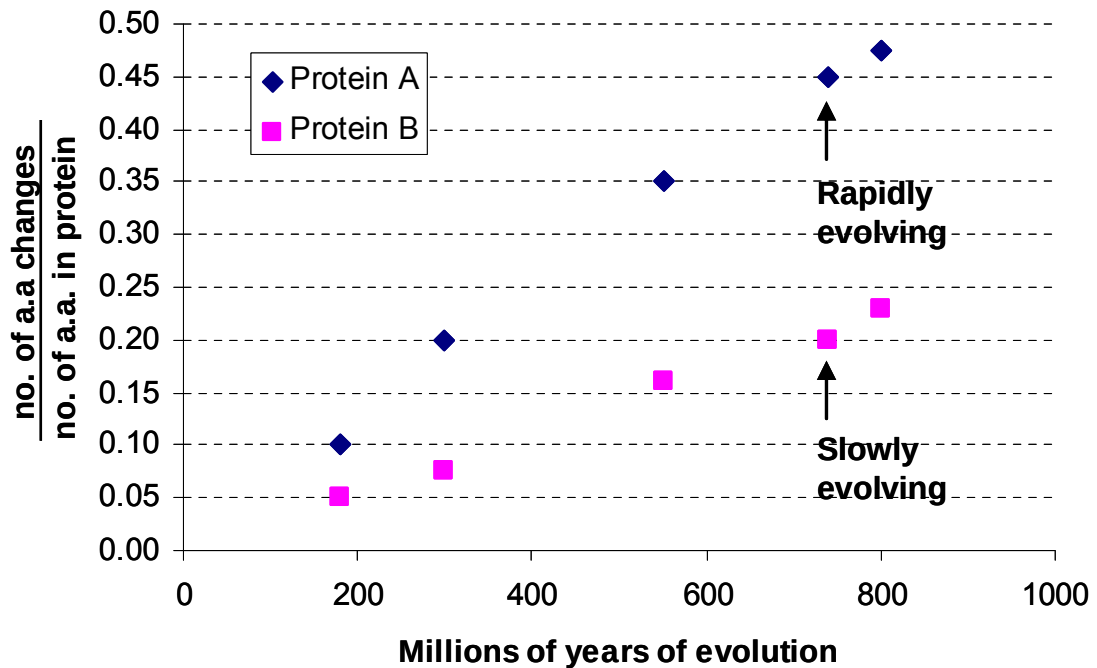
The phylogenetic tree below is based on differences between the cytochrome b proteins.



Which of the following combinations are correct?

	W	X	Y	Z
A	Lungfish	Coelacanth	Ostrich	Elephant
B	Lungfish	Ostrich	Coelacanth	Elephant
C	Coelacanth	Lungfish	Ostrich	Elephant
D	Coelacanth	Lungfish	Elephant	Ostrich

34. The graph below shows the evolution of two different proteins against the evolutionary time that has passed.

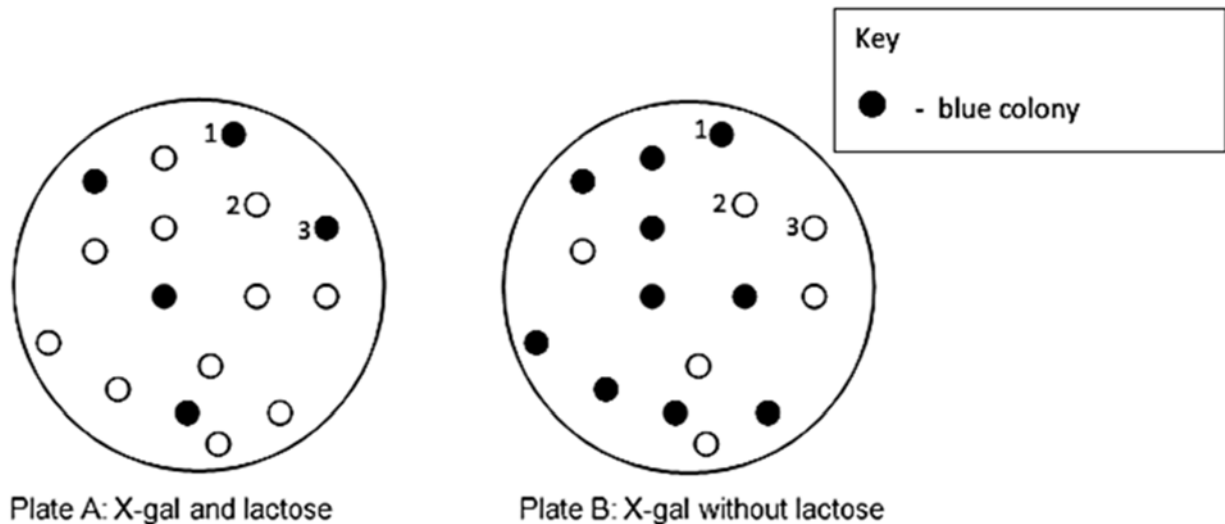


Which of the following statements can be deduced from the graphical data?

- A The difference between the amino acid sequences of protein A and protein B shows how much evolution has happened in the 800 million years.
- B The evolution of protein A is by natural selection while that of protein B is mostly neutral changes that make no difference to how the protein works.
- C Protein A has a higher proportion of possible changes that are neutral and hence evolved at a higher rate.
- D Protein B has a higher proportion of possible changes that are neutral and hence evolved at a slower rate.

35. *Escherichia coli* bacteria are infected with laboratory-cultured lambda phage. The bacteria are initially cultured in a nutrient medium without X-gal. The bacteria colonies produced are replica plated onto two agar plates, one containing X-gal and lactose and the other containing X-gal without lactose.

There is no glucose in either plates. The agar plates below show the results of this experiment.

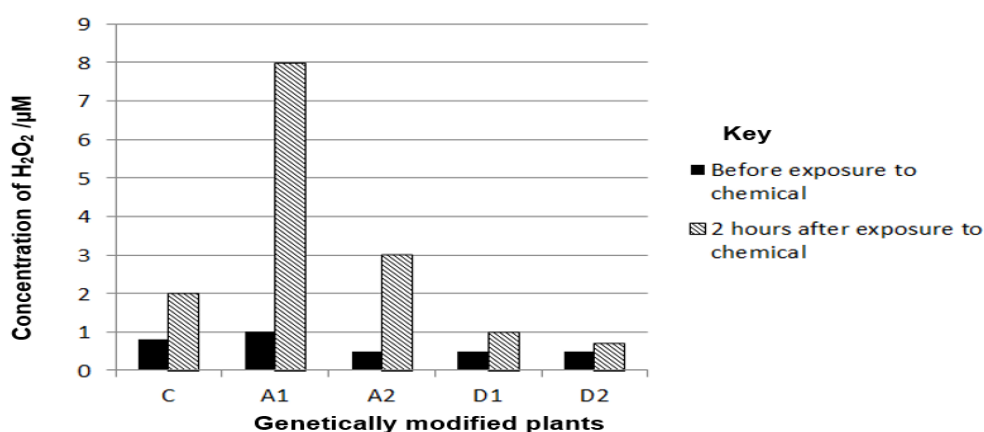


Which of the following explanations for colonies 1, 2 and 3 are correct?

- A Colony 1 is blue in both plates because transcription of Lac Z gene is turned on all the time so β galactosidase is continuously translated to break down X-gal into a blue compound.
- B Colony 2 is white in both plates because transcription of lac Z gene results in β galactosidase being produced to break down X-gal into a white compound.
- C Colony 3 is blue in plate A and white in plate B because viral DNA is integrated into lac Z gene and lac Z gene is disrupted leading to insertional inactivation.
- D Colony 3 is blue in plate A and white in plate B because phage DNA is integrated into the operator by transduction and repressor cannot find to operator.

36. Plants have developed defence mechanisms against pathogens such as bacteria, fungi and viruses. Chemicals released by these pathogens can trigger a defence response in infected plant cells. For example, the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and cellular chemicals eventually kills both the cell and pathogen.

The *OSRac1* gene from another plant species was isolated and introduced into a number of rice plant (*Oryza spp.*) lines to study its role in disease resistance of plants to blast fungus. Experiments were carried out to see if the *OSRac1* gene was part of the signalling pathway for hydrogen peroxide production. A control (C) and four other genetically modified rice plant lines (A1, A2, D1 and D2) grown *in vitro* from calluses were exposed to chemicals known to initiate a defence response by producing hydrogen peroxide. A1 and A2 are rice plants with the *OSRac1* gene always turned on. D1 and D2 are rice plants with the *OSRac1* gene suppressed. The results are shown in the graph below.



Which of the following statements can be concluded from the graph?

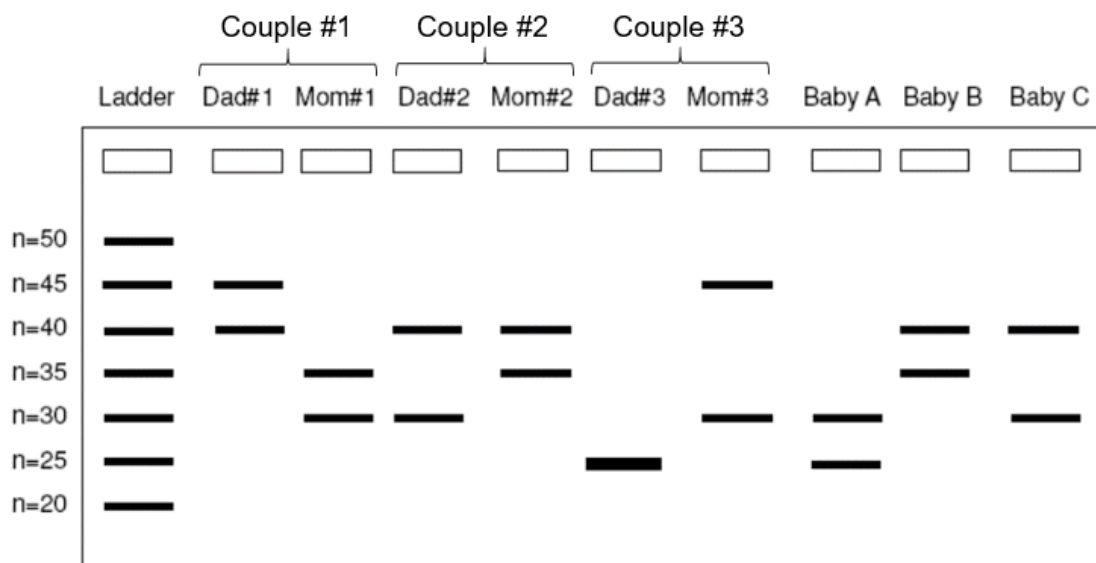
- A *OSRac1* gene is not involved in disease resistance as both D2 showed a lower increase in H_2O_2 production by 40% as compared to control which showed an increase in H_2O_2 production of 150%.
- B *OSRac1* gene is involved in disease resistance as A2 showed a higher increase in H_2O_2 production by 300% as compared to control which showed an increase in H_2O_2 production of 50%.
- C *OSRac1* gene is not involved in disease resistance as both A1 and A2 genetically modified plants showed lesser change in the number of times of H_2O_2 production.
- D *OSRac1* gene is involved in disease resistance as both D1 and D2 genetically modified plants with *OSRac1* gene suppressed showed smaller change in the number of times of H_2O_2 production.

37. In a maternity ward at a local hospital, a mix-up involving three couples and three babies caused a lot of confusion. Based on phenotypic characteristics, the nurses were unable to correctly identify the parents of the babies. In order to solve the case, a scientist was called in to carry out a DNA test to identify the parents of the babies. The test was based on the principle that different individuals have a different number of repeating units at a particular locus in a chromosome.

Chromosome 13 was isolated from the DNA samples that were obtained from the three couples and three babies and used for further analysis. The sequence below shows a segment of chromosome 13, which was used in the analysis where (TTAGGAT) is the repeating unit and n is the number of repeats.

5' ...GCTAAGTATTGCTCAAGA... (TTAGGAT) n ...GATAAATAACTGGCTAGTA...-3'
 3' ...CGATTCATAACGAGTTCT... (AATCCTA) n ... CTATTTATTGACCGATCAT...-5'

The diagram below shows the results of the DNA test obtained from each individual.



Based on the results above, which couple does Baby B belong to?

- A Couple 1
- B Couple 2
- C Couple 3
- D Not enough information

38. The DNA sequences of the normal and mutated versions of a gene are shown below.

Normal DNA sequence:

GAGAATCCTTGAGCTCTTAAGCTTATT

Mutated DNA sequence:

GAGAATCCTTGAGGTCTTAAGCTTATT

The table below shows the recognition sequences of four restriction endonucleases.

Restriction endonuclease	Recognition site
<i>Bam</i> HI	GGATCC
<i>Eco</i> RI	GAATTC
<i>Hind</i> III	AAGCTT
<i>Sac</i> I	GAGCTC

Which of the restriction endonucleases would produce different number of fragments when used to digest normal and mutant DNA?

- A *Bam*HI
 - B *Eco*RI
 - C *Hind*III
 - D *Sac*I
39. Which of the following is not true of adult stem cells during tissue repair?
- A The stem cells must have active telomerase.
 - B The different checkpoints in the cell cycle of the stem cells are activated.
 - C Mitosis of the stem cells is induced without any stimulus.
 - D The stem cells will stop dividing after the damaged cells are replaced.

40. Equal masses of tobacco plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin.

Which of the following combinations is not possible?

Treatment	Concentration of plant growth regulators / mgdm ⁻³		Effect of plant growth regulators on callus growth
	Auxin	Cytokinin	
A	2.00	0.00	No growth
B	2.00	0.50	Growth of roots
C	2.00	2.00	Increased growth of callus with no differentiation
D	2.00	3.50	Growth of roots

End of Paper